

HANZHONG, China

WMO: 571270

Lat: 33.0677N

Lon: 107.0403E

Elev: 1670

StdP: 13.83

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 28.0	(c) 29.7	(d) 14.5	(e) 12.2	(f) 41.0	(g) 19.1	(h) 15.3	(i) 39.1	(j) 10.0	(k) 38.3	(l) 8.9	(m) 37.7	(n) 1.9	(o) 200	(p) 0.333

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 13.2	(c) 93.1	(d) 76.5	(e) 90.8	(f) 75.8	(g) 88.7	(h) 75.1	(i) 79.7	(j) 88.1	(k) 78.5	(l) 86.7	(m) 77.4	(n) 85.1	(o) 4.5	(p) 90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
77.6	153.0	83.9	76.4	146.5	82.9	75.2	140.9	82.1	45.0	88.1	43.6	87.1	42.4	85.3	84.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 9.9	(b) 8.0	(c) 7.0	(e) 24.0	(f) 96.6	(g) 2.0	(h) 2.5	(i) 22.5	(j) 98.4	(k) 21.4	(l) 99.8	(m) 20.2	(n) 101.2	(o) 18.7	(p) 103.0		
(a) 9.9	(b) 8.0	(c) 7.0	(e) 22.7	(f) 81.4	(g) 2.3	(h) 1.5	(i) 21.0	(j) 82.5	(k) 19.7	(l) 83.4	(m) 18.5	(n) 84.2	(o) 16.8	(p) 85.3		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	60.4	38.7	43.7	52.7	62.5	69.7	76.0	80.1	79.3	70.3	60.7	49.8	40.6
	DBStd	15.35	3.97	4.88	6.84	6.62	5.28	4.43	4.37	5.10	5.42	5.31	5.50	3.98
	HDD50	953	351	184	50	3	0	0	0	0	0	1	72	292
	HDD65	3316	816	597	383	122	19	0	0	0	12	153	457	757
	CDD50	4759	0	7	135	379	609	779	933	909	610	334	64	0
	CDD65	1647	0	0	3	47	163	329	468	445	172	20	0	0
	CDH74	13662	0	0	25	337	1076	2540	4552	4193	881	58	0	0
	CDH80	4942	0	0	1	64	284	866	1822	1661	239	5	0	0
Wind	WSAvg	2.9	2.5	2.8	3.2	3.2	3.2	3.1	3.2	3.2	2.8	2.5	2.3	2.2
Precipitation	PrecAvg	35.4	0.3	0.4	1.3	2.8	3.8	3.9	7.0	4.9	6.5	3.0	1.3	0.3
	PrecMax	57.6	1.3	1.5	3.4	8.6	10.9	14.6	15.0	17.8	14.2	7.1	3.3	1.3
	PrecMin	24.7	0.0	0.0	0.2	0.5	0.7	0.6	1.3	0.7	1.1	0.4	0.3	0.0
	PrecStd	7.6	0.3	0.4	0.7	1.6	2.2	2.7	3.3	3.7	3.9	1.8	0.7	0.3
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	54.7	63.3	77.2	85.6	89.8	92.7	96.2	96.1	90.9	79.4	66.4	55.3
		MCWB	44.2	49.9	59.8	65.9	68.8	72.3	78.0	77.6	75.0	65.4	57.0	46.4
	2%	DB	50.8	58.6	72.0	81.6	86.1	89.9	93.3	93.0	86.2	75.0	62.9	51.5
		MCWB	42.1	47.7	56.8	63.7	67.7	72.7	77.5	77.2	72.5	64.1	55.5	44.6
	5%	DB	47.7	54.8	68.0	78.0	83.1	87.7	90.9	90.6	82.3	71.7	59.9	49.1
		MCWB	40.9	45.9	55.0	62.4	67.0	72.5	77.1	76.7	70.8	63.2	54.1	43.5
	10%	DB	45.4	51.7	63.9	74.1	80.3	85.3	88.6	88.2	78.8	68.6	57.4	47.1
		MCWB	39.9	44.6	53.0	61.2	66.1	71.9	76.3	76.1	69.7	62.5	53.6	42.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	45.5	51.9	62.5	69.5	72.7	78.4	81.7	81.5	78.1	68.8	59.3	48.7
		MCDB	51.9	59.6	74.2	80.0	82.2	84.7	89.9	90.3	86.6	72.5	62.8	51.8
	2%	WB	43.4	49.1	58.5	66.7	70.9	76.5	80.0	79.7	75.1	66.5	57.3	46.7
		MCDB	48.6	55.7	68.0	76.6	80.2	83.8	88.2	88.4	80.8	71.3	60.5	49.4
	5%	WB	42.0	47.3	56.2	64.5	69.3	74.9	78.8	78.5	72.8	65.1	55.7	45.1
		MCDB	46.6	52.7	65.5	74.0	78.3	83.1	87.2	87.0	78.8	69.8	58.6	47.9
	10%	WB	40.5	45.7	54.3	62.5	67.9	73.6	77.7	77.3	71.1	63.5	54.2	43.4
		MCDB	44.5	50.4	62.5	72.1	77.0	81.6	85.6	85.3	76.6	67.7	56.9	46.2
Mean Daily Temperature Range	5% DB	MDBR	11.7	12.6	15.3	16.3	15.8	14.0	13.2	13.4	11.1	10.4	10.4	10.8
		MCDBR	16.9	20.0	23.2	23.7	22.0	19.6	17.1	17.0	17.5	15.7	15.2	14.7
		MCWBR	9.4	9.9	9.4	8.6	7.6	7.0	5.7	5.5	6.3	6.5	8.2	8.5
	5% WB	MCDBR	15.1	16.6	20.6	19.9	18.0	15.7	14.7	14.8	13.8	12.8	11.9	11.5
MCWBR		8.9	8.8	8.9	8.8	7.5	6.7	5.7	5.6	5.6	5.8	7.1	7.3	
Clear-Sky Solar Irradiance	taub	0.622	0.701	0.721	0.693	0.650	0.633	0.649	0.689	0.663	0.657	0.648	0.588	
	taud	1.540	1.441	1.446	1.531	1.663	1.741	1.738	1.639	1.662	1.604	1.565	1.618	
	Ebn at Noon	174	177	190	207	218	221	217	205	201	186	167	174	
	Edh at Noon	72	88	94	90	79	73	73	79	75	73	70	63	
All-Sky Solar Radiation	RadAvg	675	743	1075	1369	1463	1499	1468	1353	961	801	657	620	
	RadStd	97	147	108	171	160	136	234	183	131	153	114	104	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only		+0.72	+0.92	N/A	N/A	N/A	N/A	-99	-207	+191
(47) Regional (6 neighbors)		+0.49	N/A	N/A	N/A	N/A	N/A	-72	-162	+151

Nomenclature: See separate page